⌈Thm. Let $\{a_n\}$ be a sequence of Complex Numbers.
If the series $\sum_{n=1}^{\infty} a_n$ Converges absolutely, then for any permutation $\sigma : \mathbb{N} \to \mathbb{N}$, the rearrangement $\sum_{n=1}^{\infty} a_{\sigma(n)}$ Converges the Same value.

proof Given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ s.t $m > n \geq N$ implies $\sum_{k=n}^{m} |a_k| < \varepsilon$
Take $S_j = \sum_{i=1}^{j} a_i$ and $S'_j = \sum_{i=1}^{j} a_{\sigma(i)}$.

Take $M \in \mathbb{N}$ s.t $\{1, 2, \cdots, N-1\} \subseteq \{\sigma(1), \cdots, \sigma(M)\}$ ---(*)
Then, clearly $M \geq N$, and for any $j > M$, $|S_j - S'_j| < 2\varepsilon$, so proved.

(More specifically, the (*) is obtained by:
Take $M \in \mathbb{N}$ s.t $M = \max\{\sigma^{-1}(1), \cdots, \sigma^{-1}(N-1)\}$.

Then, for any $1 \leq i \leq N-1$, $\sigma^{-1}(i) \leq M$, or $\sigma^{-1}(i) \in \{1, \cdots, M\}$.
Now, $(\sigma \circ \sigma^{-1})(i) = i \in \sigma[\{1, \cdots, M\}]$.)

But, if $\sum a_n$ does <u>Not</u> Converges absolutely, the Convergent value cannot be preserved; in actually, the Convergentness Can be broken.

Thm. Let $\sum a_n$ be a series of real numbers which converges, but not absolutely.
For any $-\infty \le \alpha \le \beta \le \infty$, there exists a rearrangement $\sum a_n'$ s.t

$$\liminf_{n\to\infty} \sum_{k=0}^{n} a_k' = \alpha \quad \text{and} \quad \limsup_{n\to\infty} \sum_{k=0}^{n} a_k' = \beta.$$

Proof. For each $n \ge 1$, let

$$p_n = \frac{|a_n| + a_n}{2}, \quad q_n = \frac{|a_n| - a_n}{2}$$

Then, $p_n + q_n = |a_n|$, $p_n - q_n = a_n$, and $p_n, q_n \ge 0$.
Moreover, both $\sum p_n$ and $\sum q_n$ diverge, because
if both are convergent, then

$$\sum_{n=0}^{N} |a_n| = \sum_{n=0}^{N} p_n + \sum_{n=0}^{N} q_n$$

converges, it is contradiction,
if only one is convergent, then

$$\sum_{n=0}^{N} a_n = \sum_{n=0}^{N} p_n - \sum_{n=0}^{N} q_n$$

diverges, and vice versa.

Observe that: for any $n \ge 0$,

$$a_n = \begin{cases} p_n & \text{if } a_n \ge 0 \\ -q_n & \text{if } a_n \le 0 \end{cases}$$

Let $\alpha_n = \frac{\alpha}{n}$, $\beta_n = \frac{\beta}{n}$ $(n \ge 2)$, $\beta_1 = |\beta|$.
Let m_1, k_1 be the smallest integers s.t

$$\begin{cases} p_1 + \cdots + p_{m_1} > \beta_1 \\ p_1 + \cdots + p_{m_1} - (q_1 + \cdots + q_{k_1}) < \alpha_1 \end{cases}$$

Let m_2, k_2 be the smallest integers s.t

$$\begin{cases} p_1 + \cdots + p_{m_1} - (q_1 + \cdots + q_{k_1}) + p_{m_1+1} + \cdots + p_{m_2} > \beta_2 \\ p_1 + \cdots + p_{m_1} - (q_1 + \cdots + q_{k_1}) + p_{m_1+1} + \cdots + p_{m_2} - (q_{k_1+1} + \cdots + q_{k_2}) < \alpha_2. \end{cases}$$

Continue in this way, and let x_n, y_n be partial sums as: $(m_0 = 0)$

$$x_n = \sum_{i=1}^{n-1} \left(\sum_{k=m_{i-1}+1}^{m_i} p_k - \sum_{k=m_{i-1}+1}^{m_i} q_k \right) + p_{m_{n-1}+1} + \cdots + p_{m_n} > \beta_n$$

$$y_n = \sum_{i=1}^{n} \left(\sum_{k=m_{i-1}+1}^{m_i} p_k - \sum_{k=m_{i-1}+1}^{m_i} q_k \right) < \alpha_n$$

Then, $|x_n - \beta_n| \le p_{m_n}$ and $|y_n - \alpha_n| \le q_{m_n}$. (Because $x_n - p_{m_n} < \beta_n$)